

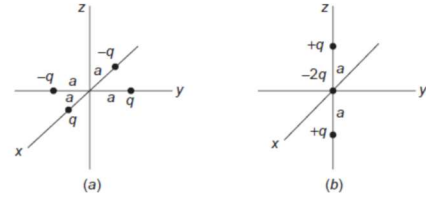
Homework 4

Electromagnetism - PHYS 715

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4.1 Calculate the multipole moments q^l_m of the charge distributions shown as parts a and b. Try to obtain results for the nonvanishing moments valid for all l , but in each case find the first two sets of nonvanishing moments at the very least.

- (c) For the charge distribution of the second set b write down the multipole expansion for the potential. Keeping only the lowest-order term in the expansion, plot the potential in the x-y plane as a function of distance from the origin for distances greater than a .
- (d) Calculate directly from Coulomb's law the exact potential for b in the x-y plane. Plot it as a function of distance and compare with the result found in part c.



Divide out the asymptotic form in parts c and d to see the behavior at large distances more clearly.

We start by investigating the term q_{lm} :-

$$q_{lm} = \int Y_{lm}^*(\theta', \phi') r'^l \underbrace{\rho(\vec{r}') d^3x'}_{\text{charge density}}$$

We'll need to express both parts a & b as charge distributions $\rho(\vec{r})$.

We are currently working with point charges so we need to use the 3D-Dirac Delta functions.

For points not on the poles $\theta \neq 0, \pi$:-

$$\delta^3(\vec{r} - \vec{r}_0) = \frac{\delta(r-r_0) \delta(\theta-\theta_0) \delta(\phi-\phi_0)}{|J(\vec{r}_0)|} = \frac{1}{r^2 \sin \theta} \delta(r-r_0) \delta(\theta-\theta_0) \delta(\phi-\phi_0) \quad (1)$$

$\uparrow$ Jacobian

For points on the poles $\theta = 0, \pi$, $\sin \theta \rightarrow 0$ which causes problems to the above form.

We can therefore use a different form working with $\mu = \cos \theta$ rather than θ .

$$d^3x = r^2 \sin \theta dr d\theta d\phi \longrightarrow d^3x = r^2 dr d\mu d\phi$$

$$\therefore \delta^3(\vec{r} - \vec{r}_0) = \frac{1}{r^2} \delta(r-r_0) \delta(\cos \theta - \cos \theta_0) \delta(\phi - \phi_0)$$

At the poles, ϕ is irrelevant and so we can drop the $\delta(\phi - \phi_0)$ term.

However, to insure that $\int \delta^3(\vec{r} - \vec{r}_0) d^3x = 1$, we must divide by a factor of 2π .

$$\therefore \delta^3(\vec{r} - \vec{r}_0) = \frac{1}{2\pi r^2} \delta(r-r_0) \delta(\cos \theta - \cos \theta_0) \quad (2)$$

$$\int_0^{2\pi} \int_{-1}^1 \int_0^\infty \frac{1}{2\pi r^2} \delta(r-r_0) \delta(\cos \theta - \cos \theta_0) r^2 dr d\mu d\phi = 1$$

For points at the origin $r=0$, θ becomes irrelevant as well and thus we need to divide by

an extra factor of $\frac{1}{2}$. $\int_{-1}^1 \int_0^{2\pi} \frac{1}{4\pi r^2} \delta(r) r^2 dr d\mu d\phi = 1$.

$$\therefore \delta^3(\vec{r} - \vec{r}_0) = \frac{1}{4\pi r^2} \delta(r) \quad (3)$$

We now are ready to solve for charge densities $\rho_a(\vec{x})$ and $\rho_b(\vec{x})$.

$$\begin{aligned} a) \rho_a(\vec{x}) &= q_{r=a, \theta=\pi/2, \phi=0} + q_{r=a, \theta=\pi/2, \phi=\pi/2} - q_{r=a, \theta=\pi/2, \phi=\pi} - q_{r=a, \theta=\pi/2, \phi=3\pi/2} \\ &= \frac{q}{a^2 \sin \theta} \delta(r-a) \delta(\theta-\pi/2) [\delta(\phi) + \delta(\phi-\pi/2) - \delta(\phi-\pi) - \delta(\phi-3\pi/2)] \quad \text{using (1)} \end{aligned}$$

$$\begin{aligned} b) \rho_b(\vec{x}) &= q_{r=a, \theta=0} - 2q_{r=0} + q_{r=a, \theta=\pi} \\ &= \frac{q}{2\pi a^2} \delta(r-a) \delta(\cos \theta - 1) - \frac{2q}{4\pi r^2} \delta(r) + \frac{q}{2\pi a^2} \delta(r-a) \delta(\cos \theta + 1) \quad \text{using (2) \& (3)} \end{aligned}$$

Let's now find q_{lm} for $\rho_a(\vec{x})$.

$$\begin{aligned} q_{lm} &= \int \int \int Y_{lm}^*(\theta', \phi') r'^l \rho_a(\vec{x}') d^3x' \\ &= \int_0^{2\pi} \int_0^\pi \int_0^\infty Y_{lm}^*(\theta', \phi') r'^l \frac{q}{a^2 \sin \theta'} \delta(r'-a) \delta(\theta'-\pi/2) [\delta(\phi') + \delta(\phi'-\pi/2) - \delta(\phi'-\pi) - \delta(\phi'-3\pi/2)] r'^2 \sin \theta' dr' d\theta' d\phi' \\ &= qa^l \int_0^{2\pi} \int_0^\pi Y_{lm}^*(\theta', \phi') \delta(\theta'-\pi/2) [\delta(\phi') + \delta(\phi'-\pi/2) - \delta(\phi'-\pi) - \delta(\phi'-3\pi/2)] d\theta' d\phi' \\ &= qa^l [Y_{lm}^*(\pi/2, 0) + Y_{lm}^*(\pi/2, \pi/2) - Y_{lm}^*(\pi/2, \pi) - Y_{lm}^*(\pi/2, 3\pi/2)] \end{aligned}$$

Using the identity $Y_{lm}^*(\theta, \phi) = \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(\cos \theta) e^{-im\phi}$

$$q_{lm} = qa^l \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(0) [1 + (-i)^m - (-1)^m - (i)^m]$$

$$= qa^l \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(0) [[1 - (-1)^m][1 - i^m]]$$

$$q_{lm} = \begin{cases} 2qa^l \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(0) (1-i^m) & m \text{ odd, } l \text{ odd} \\ 0 & m \text{ odd, } l \text{ even} \\ 0 & m \text{ even} \end{cases} \quad \begin{aligned} P_l^m(0) &= 0 \\ 1 - (-1)^m &= 0 \end{aligned}$$

We can also write an expression for $q_{l,-m}$ using the identity $P_l^{-m}(x) = (-1)^m \frac{(l-m)!}{(l+m)!} P_l^m(x)$

$$q_{l,-m} = 2qa^l (-1)^m \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(0) (1-i^m)$$

Let us analyse the first two nonvanishing sets - for $q_{\ell m}$.

For $\ell = 0$ (monopole),

$$q_{00} = 0$$

m even

→ makes sense!

No total charge.

For $\ell = 1$ (dipole),

$$q_{1,1} = qa \sqrt{\frac{3}{\pi} \frac{1}{2!}} (-1)(1-i) = qa \sqrt{\frac{3}{2\pi}} (-1+i)$$

$$q_{1,0} = 0 \quad m \text{ even}$$

$$q_{1,-1} = qa (-1) \sqrt{\frac{3}{\pi} \frac{1}{2!}} (-1)(1+i) = qa \sqrt{\frac{3}{2\pi}} (1+i)$$

For $\ell = 2$ (quadrupole),

$$q_{2,2} = q_{2,0} = q_{2,-2} = 0 \quad m \text{ even}$$

$$q_{2,1} = q_{2,-1} = 0$$

m odd, zero.

⇒ No quadrupole moment

For $\ell = 3$ (octupole):

$$q_{3,2} = q_{3,0} = q_{3,-2} = 0 \quad m \text{ even}$$

$$q_{3,3} = qa^3 \sqrt{\frac{7}{\pi} \frac{1}{6!}} (-15)(1-i^3) = qa^3 \sqrt{\frac{35}{16\pi}} (-1-i)$$

$$q_{3,1} = qa^3 \sqrt{\frac{7}{\pi} \frac{2!}{4!}} \left(\frac{3}{2}\right)(1-i^1) = qa^3 \sqrt{\frac{21}{16\pi}} (1-i)$$

$$q_{3,-1} = qa^3 (-1)^1 \sqrt{\frac{7}{\pi} \frac{2!}{4!}} \left(\frac{3}{2}\right)(1-i^{-1}) = qa^3 \sqrt{\frac{21}{16\pi}} (-1-i)$$

$$q_{3,-3} = qa^3 (-1)^3 \sqrt{\frac{7}{\pi} \frac{1}{6!}} (-15)(1-i^{-3}) = qa^3 \sqrt{\frac{35}{16\pi}} (1-i)$$

$$P_\ell^m(x) = (-1)^m (1-x^2)^{m/2} \frac{d^m}{dx^m} P_\ell(x), \quad P_\ell^m(x) = (-1)^m \frac{(\ell-m)!}{(\ell+m)!} P_\ell^m(x).$$

$$\ell = 0$$

$$P_0^0 = 1$$

$$\ell = 1$$

$$P_1^0 = x, \quad P_1^1 = -\sqrt{1-x^2}$$

$$\ell = 2$$

$$P_2^0 = \frac{1}{2}(3x^2 - 1),$$

$$P_2^1 = -3x\sqrt{1-x^2},$$

$$P_2^2 = 3(1-x^2)$$

$$\ell = 3$$

$$P_3^0 = \frac{1}{2}(5x^3 - 3x),$$

$$P_3^1 = -\frac{1}{2}(15x^2 - 3)\sqrt{1-x^2} = -\frac{3}{2}(5x^2 - 1)\sqrt{1-x^2},$$

$$P_3^2 = 15x(1-x^2),$$

$$P_3^3 = -15(1-x^2)^{3/2}$$

b) Recall $\rho_b(\vec{r}) = \frac{q}{2\pi a^2} \delta(r-a) [\delta(\cos\theta - 1) - \frac{2q}{4\pi a^2} \delta(r) + \frac{q}{2\pi a^2} \delta(r-a) \delta(\cos\theta + 1)]$

Plugging in ρ_b in the definition of q_{lm}

$$\begin{aligned} q_{lm} &= \int Y_{lm}^*(\theta', \phi') r'^l \rho_b(\vec{r}') d\vec{r}' \\ &= \int_0^{2\pi} \int_0^\pi \int_0^\infty Y_{lm}^*(\theta', \phi') r'^l \left[\frac{q}{2\pi a^2} \delta(r'-a) [\delta(\cos\theta' - 1) + \delta(\cos\theta' + 1)] - \frac{2q}{4\pi r'^2} \delta(r') \right] r'^2 \sin\theta' dr' d\theta' d\phi' \\ &= \underbrace{\int_0^{2\pi} \int_0^\pi Y_{lm}^*(\theta', \phi') a^l \frac{q}{2\pi} (\delta(\cos\theta' - 1) + \delta(\cos\theta' + 1)) \sin\theta' d\theta' d\phi'}_{I_1} - \underbrace{\int_0^{2\pi} \int_0^\pi \int_0^\infty Y_{lm}^*(\theta', \phi') r'^l \frac{2q}{4\pi} \delta(r') \sin\theta' dr' d\theta' d\phi'}_{I_2} \end{aligned}$$

For I_1 ,

$$\begin{aligned} I_1 &= \frac{qa^l}{2\pi} \int_0^{2\pi} Y_{lm}^*(0, \phi') + Y_{lm}^*(\pi, \phi') d\phi' \\ &= \frac{qa^l}{2\pi} \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} \underbrace{[P_l^m(-1) + P_l^m(1)]}_{\delta_{m0} + (-1)^l \delta_{m0}} \underbrace{\int_0^{2\pi} e^{-im\phi'} d\phi'}_{2\pi \delta_{m0}} \\ &= \begin{cases} qa^l \sqrt{\frac{2l+1}{4\pi}} [1 + (-1)^l] & \text{for } m=0 \\ 0 & \text{for } m \neq 0 \end{cases} \end{aligned}$$

For I_2 ,

$$\begin{aligned} I_2 &= \int_0^{2\pi} \int_0^\pi \int_0^\infty Y_{lm}^*(\theta', \phi') r'^l \frac{2q}{4\pi} \delta(r') \sin\theta' dr' d\theta' d\phi' \\ &= \frac{2q}{4\pi} \int_0^{2\pi} \int_0^\pi \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} \delta_{l0} P_l^m(\cos\theta') \sin\theta' e^{-im\phi'} d\theta' d\phi' \\ &= \frac{2q}{4\pi} \int_0^\pi \underbrace{\sqrt{\frac{2l+1}{4\pi}} \frac{(l-m)!}{(l+m)!}}_{\frac{1}{4\pi}} \underbrace{\delta_{l0} P_l^m(\cos\theta') \sin\theta' d\theta'}_{P_0^0(x) = 1} \underbrace{\int_0^{2\pi} e^{-im\phi'} d\phi'}_{2\pi \delta_{m0}} \\ &= \frac{2q}{4\pi} \cdot \sqrt{\frac{1}{4\pi}} \delta_{l0} \int_{-1}^1 P_0^0(x) dx \cdot 2\pi \delta_{m0} \\ &= q \sqrt{\frac{1}{\pi}} \delta_{l0} \delta_{m0} \end{aligned}$$

This causes odd l terms to vanish!

Therefore, $q_{lm} = qa^l \sqrt{\frac{2l+1}{4\pi}} [1 + (-1)^l] \delta_{m0} - q \sqrt{\frac{1}{\pi}} \delta_{m0} \delta_{l0}$

Let's now find the nonvanishing terms for q_{lm} .

$$l=0 \quad q_{l00} = q \sqrt{\frac{1}{4\pi}} (2) - q \sqrt{\frac{1}{\pi}} = 0$$

$$l=1 \quad q_{1,1} = q_{1,0} = q_{1,-1} = 0 \quad \text{odd } l$$

$$l=2 \quad q_{2,0} = q a^2 \sqrt{\frac{5}{\pi}} \quad q_{2,m} = 0 \quad \text{for } m \neq 0$$

$$l=3 \quad \text{All } q_{3,m} = 0 \quad \text{odd } l.$$

$$l=4 \quad q_{4,0} = q a^4 \sqrt{\frac{9}{\pi}} \quad q_{4,m} = 0 \quad \text{for } m \neq 0$$

$$\Rightarrow q_{lm} = \begin{cases} q a^l \sqrt{\frac{2l+1}{\pi}} & \text{for } m=0, l \text{ even} \\ 0 & \text{otherwise} \end{cases}$$

c) for part c, we are asked to find an approximation for Φ using the multipole expansion for set up b.

Writing the multipole expansion of Φ :

$$\Phi = \frac{1}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{4\pi}{2l+1} q_{lm} \frac{Y_{lm}(\theta, \phi)}{r^{l+1}}$$

Plugging in the value of q_{lm} from part b

$$\begin{aligned} \Phi &= \frac{1}{4\pi\epsilon_0} \sum_{l=2, \text{even}}^{\infty} \frac{4\pi}{2l+1} q \frac{a^l}{r^{l+1}} \sqrt{\frac{2l+1}{\pi}} \underbrace{Y_{l0}(\theta, \phi)}_{\sqrt{\frac{2l+1}{4\pi}} P_l(\cos\theta)} \\ &= \frac{q}{2\pi\epsilon_0 a} \sum_{l=2, \text{even}}^{\infty} \left(\frac{a}{r}\right)^{l+1} P_l(\cos\theta) \end{aligned}$$

If $a \ll r$, then $\frac{a}{r} \ll 1$ and thus we can drop higher order $\left(\frac{a}{r}\right)$ terms.

$$\Phi \approx \frac{q}{2\pi\epsilon_0 a} \frac{a^3}{r^3} P_2(\cos\theta) = \frac{q}{4\pi\epsilon_0 a} \left(\frac{a}{r}\right)^3 (3\cos^2\theta - 1)$$

To plot in the x - y plane, we take $\theta = \pi/2$.

$$\Phi = \underbrace{\frac{-q}{4\pi\epsilon_0 a}}_{\text{units of } \phi} \underbrace{\left(\frac{1}{(r/a)^3}\right)}_x$$

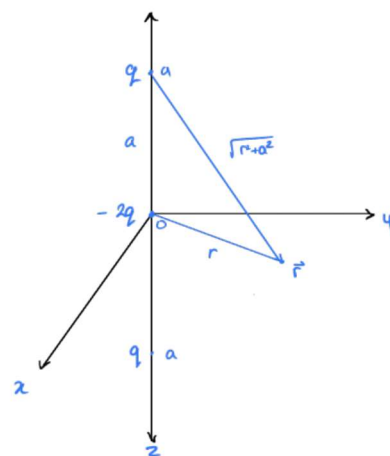
d) For this problem, we can actually find the exact potential using Coulomb's Law.
We compare the exact potential with our approximation in part c.

$$\Phi = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{|\vec{r} + a\hat{z}|} + \frac{1}{|\vec{r} - a\hat{z}|} - \frac{2}{|\vec{r}|} \right]$$

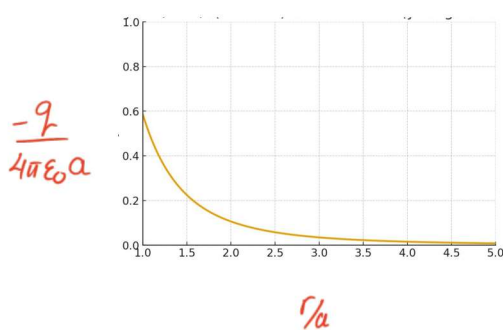
Looking at the solution in the x - y plane is

$$\begin{aligned} \Phi &= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{\sqrt{r^2 + a^2}} + \frac{1}{\sqrt{r^2 + a^2}} - \frac{2}{r} \right] \\ &= \frac{q}{4\pi\epsilon_0 a} \left[\frac{2a}{\sqrt{r^2 + a^2}} - \frac{2a}{r} \right] \\ &= \frac{-q}{4\pi\epsilon_0 a} \left[\frac{2}{\underbrace{(r/a)}_x} - \frac{2}{\underbrace{\sqrt{(r/a)^2 + 1}}_x} \right] \end{aligned}$$

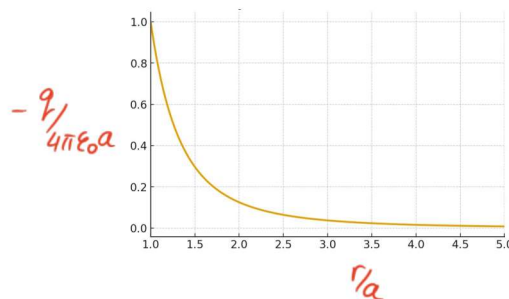
units of y



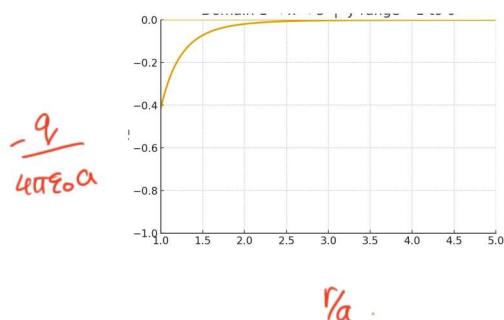
Approximate Φ from (c)



Exact Φ from (d)



By finding the difference $\Phi_{\text{exact}} - \Phi_{\text{app}}$, we can conclude



We see that the first order approximation does well beyond $r > 2a$. Otherwise, it overshoots the true potential for $r < 2a$.

4.2 A point dipole with dipole moment $\mathbf{p}$ is located at the point $\mathbf{x}_0$. From the properties of the derivative of a Dirac delta function, show that for calculation of the potential Φ or the energy of a dipole in an external field, the dipole can be described by an effective charge density

$$\rho_{\text{eff}}(\mathbf{x}) = -\mathbf{p} \cdot \nabla \delta(\mathbf{x} - \mathbf{x}_0)$$

We work with the potential Φ of the dipole to find ρ_{eff}

The potential due to a dipole is

$$\Phi = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot (\vec{x} - \vec{x}_0)}{|\vec{x} - \vec{x}_0|^3} \quad (1.25)$$

Using the Poisson equation $\nabla^2 \Phi = -\frac{\rho}{\epsilon_0}$, we find ...

$$\begin{aligned} \rho_{\text{eff}} &= -\epsilon_0 \nabla^2 \Phi = -\frac{1}{4\pi} \vec{p} \cdot \nabla^2 \left(\frac{\vec{x} - \vec{x}_0}{|\vec{x} - \vec{x}_0|^3} \right) = \frac{1}{4\pi} \vec{p} \cdot \nabla^2 \left(\frac{1}{|\vec{x} - \vec{x}_0|} \right) \\ &= \frac{1}{4\pi} \vec{p} \cdot \nabla \underbrace{\nabla^2 \left(\frac{1}{|\vec{x} - \vec{x}_0|} \right)}_{-4\pi \delta(\vec{x} - \vec{x}_0)} \\ &= -\vec{p} \cdot \nabla \delta(\vec{x} - \vec{x}_0) \end{aligned}$$

4.5 A localized charge density $\rho(x, y, z)$ is placed in an external electrostatic field described by a potential $\Phi^{(0)}(x, y, z)$. The external potential varies slowly in space over the region where the charge density is different from zero.

- (a) From first principles calculate the total force acting on the charge distribution as an expansion in multipole moments times derivatives of the electric field, up to and including the quadrupole moments. Show that the force is

$$\mathbf{F} = q\mathbf{E}^{(0)}(0) + \left\{ \nabla [\mathbf{p} \cdot \mathbf{E}^{(0)}(\mathbf{x})] \right\}_0 + \left\{ \nabla \left[\frac{1}{6} \sum_{j,k} Q_{jk} \frac{\partial E_j^{(0)}}{\partial x_k}(\mathbf{x}) \right] \right\}_0 + \dots$$

Compare this to the expansion (4.24) of the energy W . Note that (4.24) is a number—it is not a function of $\mathbf{x}$ that can be differentiated! What is its connection to $\mathbf{F}$?

- (b) Repeat the calculation of part a for the total torque. For simplicity, evaluate only one Cartesian component of the torque, say N_1 . Show that this component is

$$N_1 = [\mathbf{p} \times \mathbf{E}^{(0)}(0)]_1 + \frac{1}{3} \left[\frac{\partial}{\partial x_3} \left(\sum_j Q_{2j} E_j^{(0)} \right) - \frac{\partial}{\partial x_2} \left(\sum_j Q_{3j} E_j^{(0)} \right) \right] + \dots$$

This problem involves sum notation manipulation with 3D Taylor Series.

There is a lot of set up involved so let's get started.

* Force : $\vec{F} = \int \rho(\vec{x}) \vec{E}(\vec{x}) d^3x$

* Divergence : $\vec{\nabla} \cdot \equiv \partial_x + \partial_y + \partial_z \equiv \sum_i \partial_{x_i}$

* Taylor Series 1D : $f(x) = f(x_0) + x f'(x_0) + \frac{1}{2!} x^2 f''(x_0) + \dots$

* Taylor Series 3D :
$$f(\vec{x}) = f(\vec{x}_0) + x \partial_x f(x_0) + y \partial_y f(x_0) + z \partial_z f(x_0) + \frac{1}{2!} \left[\begin{array}{ccc} x^2 \partial_{xx} f(x_0) & + & xy \partial_{yx} f(x_0) & + & zx \partial_{zx} f(x_0) \\ xy \partial_{xy} f(x_0) & + & y^2 \partial_{yy} f(x_0) & + & zy \partial_{zy} f(x_0) \\ xz \partial_{xz} f(x_0) & + & yz \partial_{yz} f(x_0) & + & z^2 \partial_{zz} f(x_0) \end{array} \right] + \dots$$

$$= f(\vec{x}_0) + \sum_i x_i \partial_{x_i} f(x_0) + \frac{1}{2} \sum_i \sum_j x_i x_j \partial_{x_i} \partial_{x_j} f(x_0)$$

* The force is a vector field and thus we will need to do the 3D Taylor Series for each component.

* Dipole Moment : $\vec{p} = \int \vec{x}' \rho(\vec{x}') d^3x'$

* Quadrupole Moment : $Q_{ij} = \int (3x'_i x'_j - r'^2 \delta_{ij}) \rho(\vec{x}') d^3x'$

I think we are now ready to tackle this problem.

a) Writing the force and expanding the external electric field $\vec{E}(\vec{x})$ in the 3D Taylor series

$$\vec{F} = \int \rho(\vec{x}) \vec{E}(\vec{x}) d^3x$$

$$= \sum_i \hat{x}_i \int \rho(\vec{x}) E_i(\vec{x}) d^3x \quad \text{splitting the force component wise.}$$

$$E_i(\vec{x}) = \left[E_i(\vec{x}_0) + \sum_j x_j \partial_{x_j} E_i(\vec{x}_0) + \frac{1}{2} \sum_j \sum_k x_j x_k \partial_{x_j} \partial_{x_k} E_i(\vec{x}_0) + \dots \right]_{\vec{x}_0 = \vec{0}}$$

Plugging the electric field expansion in the force:

$$\vec{F} = \underbrace{\sum_i \hat{x}_i \int \rho(\vec{x}) E_i(\vec{x}_0) d^3\vec{x}}_{\text{monopole}} + \underbrace{\sum_i \hat{x}_i \int \rho(\vec{x}) \sum_j x_j \partial_{x_j} E_i(\vec{x}_0) d^3\vec{x}}_{\text{dipole}} + \underbrace{\frac{1}{2} \sum_i \hat{x}_i \int \rho(\vec{x}) \sum_j \sum_k x_j x_k \partial_{x_j} \partial_{x_k} E_i(\vec{x}_0) d^3\vec{x}}_{\text{quadrupole}} + \dots \Big|_{\vec{x}_0 = \vec{0}}$$

To further simplify the dipole & quadrupole terms, we use $\vec{\nabla} \times \vec{E} = 0$ i.e. $\partial_{x_i} E_j = \partial_{x_j} E_i$

$$= \sum_i \hat{x}_i E_i(\vec{x}_0) q + \sum_i \hat{x}_i \int \rho(\vec{x}) \sum_j \partial_{x_j} E_i(\vec{x}_0) d^3\vec{x} + \frac{1}{2} \sum_i \hat{x}_i \int \rho(\vec{x}) \sum_j \sum_k x_j x_k \partial_{x_j} \partial_{x_k} E_i(\vec{x}_0) d^3\vec{x} + \dots \Big|_{\vec{x}_0 = \vec{0}}$$

We notice that the Electric field terms with their derivatives do not depend on the variables in the integrals & so we factor them out.

$$\begin{aligned} &= \vec{E}(\vec{x}_0) q + \sum_i \hat{x}_i \partial_{x_i} \sum_j E_j(\vec{x}_0) \int x_j \rho(\vec{x}) d^3\vec{x} + \frac{1}{2} \sum_i \hat{x}_i \partial_{x_i} \sum_j \sum_k \partial_{x_k} E_j(\vec{x}_0) \int \rho(\vec{x}) x_j x_k d^3\vec{x} + \dots \Big|_{\vec{x}_0 = \vec{0}} \\ &= \vec{E}(\vec{x}_0) q + \underbrace{\sum_i \hat{x}_i \partial_{x_i}}_{\vec{\nabla} \cdot} \underbrace{\sum_j E_j(\vec{x}_0)}_{\vec{p} \cdot \vec{E}} \underbrace{\int x_j \rho(\vec{x}) d^3\vec{x}}_{\text{dipole component } P_j} + \frac{1}{2} \underbrace{\sum_i \hat{x}_i \partial_{x_i}}_{\vec{\nabla} \cdot} \left[\underbrace{\sum_j \sum_k \partial_{x_k} E_j(\vec{x}_0)}_{\text{quadrupole component } \frac{Q_{jk}}{3} + \int r^2 \delta_{jk} \rho(\vec{x}) d^3\vec{x}} \right] \Big|_{\vec{x}_0 = \vec{0}} \\ &= q \vec{E}(\vec{x}_0) + \underbrace{\vec{\nabla} \cdot [\vec{p} \cdot \vec{E}]}_{\vec{\nabla} \cdot} + \frac{1}{6} \underbrace{\vec{\nabla} \cdot [\sum_j \sum_k \partial_{x_k} E_j(\vec{x}_0) Q_{jk}]}_{\vec{\nabla} \cdot} \Big|_{\vec{x}_0 = \vec{0}} \end{aligned}$$

$\vec{\nabla} \cdot \vec{E} = 0$ as there is no charge enclosed in volume of interest.
 $\vec{\nabla} \cdot \vec{E} = \sum_j \partial_{x_j} E_j = 0$
 $\sum_j \partial_{x_k} \delta_{jk} E_j = \sum_j \partial_{x_j} E_j = 0$

And we are done! Oh wait woops

one final thing...

We compare the work done expansion with what we got to find that...

$$W = q \Phi(0) - \vec{p} \cdot \vec{E}(0) - \frac{1}{6} \sum_i \sum_j Q_{ij} \frac{\partial E_i}{\partial x_j}(0) + \dots$$

Rewriting $\vec{E}(\vec{x}_0)$ as $-\nabla \Phi$ and factoring out the gradient in $\vec{F}$.

$$\vec{F} = -\nabla \left[q \Phi(0) - \vec{p} \cdot \vec{E}(0) - \frac{1}{6} \sum_i \sum_j Q_{ij} \frac{\partial E_i}{\partial x_j}(0) + \dots \right]$$

Great! Just like how we know it $\vec{F} = -\nabla W$.

We now have all the tools to deal with the Torque.

The electrostatic torque for a charge distribution ρ from an external field E is:

$$N = \int \vec{r} \times [\rho(\vec{r}) \vec{E}(\vec{r})] d^3\vec{r}$$

Jackson recommended to work with only one component and that is what we will do:

$$N_1 = \int \rho(\vec{r}) [x_2 E_3(\vec{r}) - x_3 E_2(\vec{r})] d^3\vec{r}$$

Using the first order electric field expansions for E_2 and E_3

$$E_2(\vec{r}) = E_2(\vec{r}_0) + \sum_j x_j \partial_{x_j} E_2(\vec{r}_0) + \dots \Big|_{\vec{r}_0=0}$$

$$E_3(\vec{r}) = E_3(\vec{r}_0) + \sum_j x_j \partial_{x_j} E_3(\vec{r}_0) + \dots \Big|_{\vec{r}_0=0}$$

$$N_1 = \underbrace{\int \rho(\vec{r}) x_2 E_3(\vec{r}_0) d^3\vec{r}}_{P_2} - \underbrace{\int \rho(\vec{r}) x_3 E_2(\vec{r}_0) d^3\vec{r}}_{P_3} + \int \rho(\vec{r}) \sum_j x_2 x_j \underbrace{\partial_{x_j} E_3(\vec{r}_0)}_{\frac{Q_{2j}}{3}} d^3\vec{r} - \int \rho(\vec{r}) \sum_j x_3 x_j \underbrace{\partial_{x_j} E_2(\vec{r}_0)}_{\frac{Q_{3j}}{3}} d^3\vec{r} \Big|_{\vec{r}_0=0}$$

We recall that the $\int r^2 \delta_{jk} \rho(\vec{r}) d^3\vec{r} \rightarrow 0$ in the quadrupole.

$$N_1 = P_2 E_3(\vec{r}_0) - P_3 E_2(\vec{r}_0) + \frac{1}{3} \sum_j \partial_{x_j} E_3(\vec{r}_0) Q_{2j} - \frac{1}{3} \sum_j \partial_{x_j} E_2(\vec{r}_0) Q_{3j} \Big|_{\vec{r}_0=0}$$

$$\text{Using } \partial_{x_i} E_j = \partial_{x_j} E_i \quad (\vec{\nabla} \times \vec{E} = \vec{0})$$

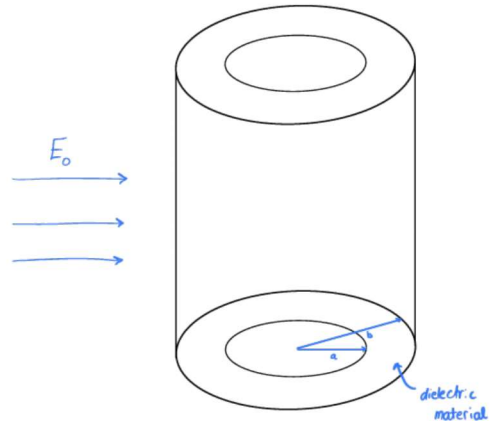
$$N_1 = [\rho \times \vec{E}(\vec{r}_0)]_1 + \frac{1}{3} \partial_{x_3} \left[\sum_j Q_{2j} E_3 - \sum_j Q_{3j} E_2 \right]$$

4.6 A very long, right circular, cylindrical shell of dielectric constant ϵ/ϵ_0 and inner and outer radii a and b , respectively, is placed in a previously uniform electric field E_0 with its axis perpendicular to the field. The medium inside and outside the cylinder has a dielectric constant of unity.

- Determine the potential and electric field in the three regions, neglecting end effects.
- Sketch the lines of force for a typical case of $b \simeq 2a$.
- Discuss the limiting forms of your solution appropriate for a solid dielectric cylinder in a uniform field, and a cylindrical cavity in a uniform dielectric.

a) This is a relatively a new shape that we need to find the potential Φ for.

We start by solving the 2D Laplace Equation as the symmetry is independent of z .



$$\nabla^2 \Phi = \frac{\partial^2 \Phi}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial \Phi}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 \Phi}{\partial \phi^2} = 0$$

We seek a separation of variables $\Phi(\rho, \phi) = R(\rho) Q(\phi)$

$$\nabla^2(RQ) = Q \frac{\partial^2 R}{\partial \rho^2} + Q \frac{1}{\rho} \frac{\partial R}{\partial \rho} + \frac{1}{\rho^2} R \frac{\partial^2 Q}{\partial \phi^2} = 0$$

$$\Rightarrow Q(R'' + \frac{1}{\rho} R') + \frac{R}{\rho^2} Q'' = 0$$

$$\Rightarrow \frac{\rho^2(R'' + \frac{1}{\rho} R')}{R} + \frac{Q''}{Q} = 0 \Rightarrow \frac{Q''}{Q} = -\frac{\rho^2(R'' + \frac{1}{\rho} R')}{R} = -\lambda$$

We continue to solve for each part.

Angular Part

$Q'' + \lambda Q = 0$ letting $\lambda = m^2$. The general solution to this ODE $\rightarrow$

$$Q(\phi) = A_m \cos m\phi + B_m \sin m\phi$$

Radial Part

$$\rho^2 R'' + \rho R' - m^2 R = 0$$

Making an educated guess of $R = \rho^k$, we have that :

$$\rho^2 \cdot k(k-1) \rho^{k-2} + \rho \cdot k \rho^{k-1} - m^2 \rho^k = 0$$

$$\rho^k [k(k-1) + k - m^2] = 0$$

The non-trivial solution then will be $k^2 - m^2 = 0 \rightarrow k = \pm m$.

$$\text{Therefore, } R(\rho) = a_m \rho^m + b_m \rho^{-m} \quad [m \geq 1]$$

We notice that $m=0$ cause issues. ($R(\rho) \rightarrow \infty$ for $m=0$)

We solve for that case separately.

$$\rho^2 R'' + \rho R' = 0$$

$$\rho R'' + R' = 0$$

$$(\rho R')' = 0 \Rightarrow \rho R' = k \quad \text{where } k \text{ is constant}$$

$$R' = \frac{k}{\rho}$$

$$R(\rho) = A + B \ln(\rho) \quad [m=0]$$

Adding now all possible solutions ...

$$\Phi(\rho, \theta) = a_0 + b_0 \ln \rho + \sum_{m=1} [a_m \rho^m + b_m \rho^{-m}] [A_m \cos m\theta + B_m \sin m\theta] \quad (1)$$

We work on making things convenient by introducing the following.

$$1) \Phi \rightarrow -E_0 \rho \cos \theta \quad \text{for } \rho \rightarrow \infty$$

proof $\vec{E} = E_0 \hat{x}$ for external electric field

$$\Phi = - \int E_0 \hat{x} \cdot d\vec{l} = - \int E_0 dx = -E_0 x = -E_0 \rho \cos \theta \quad \text{in cylindrical coordinates}$$

2) Region	Permittivity
$\rho < a, \rho > b$	1
$a < \rho < b$	$\epsilon = \epsilon_0$

let's now find the general form for potential in all 3 regions :-

This is a 2D-problem with azimuthal dependence of $\cos \theta$ i.e. $m=1$ in eqⁿ (1)

$$\therefore \Phi(\rho, \theta) = (A\rho + B\rho^{-1}) \cos \theta$$

Region I ($\rho < a$) : $\Phi_{\rho < a} = -E_1 \rho \cos \theta$

$B=0$ to keep Φ finite at the origin

Region II ($a < \rho < b$) : $\Phi_{a < \rho < b} = -(A\rho + B\rho^{-1}) \cos \theta$

Both $\rho, 1/\rho$ left in middle region

Region III ($\rho > b$) : $\Phi_{\rho > b} = -(E_0 \rho + C\rho^{-1}) \cos \theta$

Field must be uniform for $\rho \rightarrow \infty$, $\Phi \rightarrow -E_0 \rho \cos \theta$.

Electric Field Components :

In cylindrical coordinates, $E_r = -\partial_r \Phi$, $E_\theta = -\frac{1}{r} \partial_\theta \Phi$

	E_r	E_θ
Region I ($\rho < a$) :	$E_1 \cos \theta$	$-E_1 \sin \theta$
Region II ($a < \rho < b$) :	$(A - \frac{B}{\rho^2}) \cos \theta$	$-(A + \frac{B}{\rho^2}) \sin \theta$
Region III ($\rho > b$) :	$(E_0 - \frac{C}{\rho^2}) \cos \theta$	$-(E_0 + \frac{C}{\rho^2}) \sin \theta$

$\Rightarrow$ Field is uniform!

$$E_1 (\cos \theta \hat{r} - \sin \theta \hat{\theta}) = E_1 \hat{x}$$

Boundary Conditions :

There are 3 boundary conditions of interest namely :

1) Potential is continuous. $\Phi_i = \Phi_j$ at boundary

2) Tangential component of $\vec{E}$ is continuous. $(E_j - E_i) \times \hat{n} = 0 \Rightarrow E_{i\theta} = E_{j\theta}$

3) Normal component of $\vec{D}$ is continuous $(D_j - D_i) \cdot \hat{n} = \rho = 0 \Rightarrow \epsilon_j E_{j\rho} = \epsilon_i E_{i\rho}$
no free charge

* Since everything varies with $\cos \theta$ & $\sin \theta$, we can just work with amplitudes of the angular parts.

Inner Boundary ($r=a$)

I Φ continuous

$$\Phi_{\rho < a} = \Phi_{a < \rho < b}$$

$$E_r = A + \frac{B}{a^2} \quad \text{same!}$$

Outer Boundary ($r=b$)

$$\Phi_{a < \rho < b} = \Phi_{\rho > b}$$

$$A + \frac{B}{b^2} = E_0 + \frac{C}{b^2} \quad \text{same!}$$

II $\vec{E}_t$ continuous

$$E_\theta^{(1)} = E_\theta^{(2)}$$

$$E_r = A + \frac{B}{a^2} \quad \text{same!}$$

$$E_\theta^{(2)} = E_\theta^{(3)}$$

$$A + \frac{B}{b^2} = E_0 + \frac{C}{b^2} \quad \text{same!}$$

III $\vec{D}_n$ continuous

$$E_\rho^{(1)} = \kappa E_\rho^{(2)}$$

$$E_r = \kappa (A - B/a^2)$$

$$\kappa E_\rho^{(2)} = E_\rho^{(3)}$$

$$A - B/b^2 = \kappa (E_0 - C/b^2)$$

Solving the System of Equations :

We now have 4 independent equations with 4 unknowns $[E, A, B, C]$.

Using a software to solve it, we have the solutions :

Let $D_0 = b^2(x+1)^2 - a^2(x-1)^2$ convenient notation

Then, $E_r = \frac{4xb^2}{D_0} E_0$

$$A = \frac{2(x+1)b^2}{D_0} E_0$$

$$B = \frac{2(x-1)a^2b^2}{D_0} E_0$$

$$C = -\frac{(x^2-1)(b^2-a^2)b^2}{D_0} E_0$$

Determining the Potential & Electric Field :

	Potential Φ	Electric Field E_r	Electric Field E_θ
Region I ($\rho < a$) :	$-E_0 \frac{4xb^2}{D_0} \rho \cos\theta$	$E_0 \frac{4xb^2}{D_0} \cos\theta$	$-E_0 \frac{4xb^2}{D_0} \sin\theta$
Region II ($a < \rho < b$) :	$-E_0 \frac{2b^2}{D_0} \left[(x+1)\rho + (x-1)\frac{a^2}{\rho} \right] \cos\theta$	$E_0 \frac{2b^2}{D_0} \left[(x+1) - (x-1)\frac{a^2}{\rho^2} \right] \cos\theta$	$-E_0 \frac{2b^2}{D_0} \left[(x+1) + (x-1)\frac{a^2}{\rho^2} \right] \sin\theta$
Region III ($\rho > b$) :	$-E_0 \left[\rho - \frac{(x^2-1)(b^2-a^2)b^2}{D_0} \frac{1}{\rho} \right] \cos\theta$	$E_0 \left[1 - \frac{(x^2-1)(b^2-a^2)b^2}{D_0} \frac{1}{\rho^2} \right] \cos\theta$	$-E_0 \left[1 + \frac{(x^2-1)(b^2-a^2)b^2}{D_0} \frac{1}{\rho^2} \right] \sin\theta$

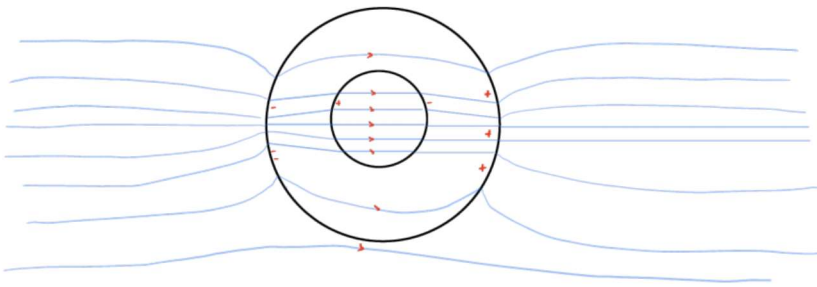
* Observation : $E_{\rho < a}$ is uniform with $\vec{E}_{\rho < a} = \frac{4xb^2}{D_0} E_0 \hat{x}$
 $E_{\rho > b}$ is uniform at $\rho \rightarrow \infty$ with $\vec{E}_{\rho > b} = E_0 \hat{x}$

b) for $b \approx 2a$,

Region I ($\rho < a$) : Uniform Electric field, parallel to E_0 but weaker

Region II ($a < \rho < b$) : Higher Electric field, non uniform, bow with curvature, continuous with both regions

Region III ($\rho > b$) : Uniform away from cylinder, distort when close to cylinder



c) Limiting Cases :

i) Solid dielectric cylinder in uniform field ($a=0$)

We shrink the inner radius to zero $\therefore a \rightarrow 0$

Therefore, we have only 2 regions. $\Phi_{\rho < b}$ & $\Phi_{\rho > b}$

for $a \rightarrow 0$, $D_0 \rightarrow b^2(\chi+1)^2$

Potential Φ

$$\text{Region I } (\rho < b) : \lim_{a \rightarrow 0} -E_0 \frac{2b^2}{D_0} \left[(\chi+1)\rho + (\chi-1)\frac{a^2}{\rho} \right] \cos\theta = -\frac{2}{\chi+1} E_0 \rho \cos\theta \rightarrow \text{uniform field reduced by } \frac{2}{\chi+1}$$

$$\text{Region II } (\rho > b) : \lim_{a \rightarrow 0} -E_0 \left[\rho - \frac{(\chi^2-1)(b^2-a^2)b^2}{D_0} \frac{1}{\rho} \right] \cos\theta = \underbrace{-E_0 \rho \cos\theta}_{\text{original field}} + \underbrace{E_0 \frac{\chi-1}{\chi+1} \frac{b^2}{\rho} \cos\theta}_{\text{Dipole perturbation near cylinder}}$$

ii) Cylindrical Cavity in dielectric material

We let the outer radius go to infinity $\therefore b \rightarrow \infty$

Therefore, we have only 2 regions. $\Phi_{\rho < a}$ & $\Phi_{\rho > a}$

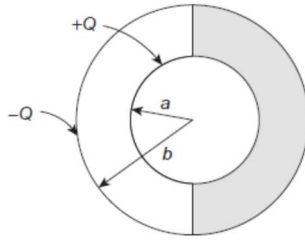
for $b \rightarrow \infty$, $D_0 \rightarrow b^2(\chi+1)^2$ dominates

Potential Φ

$$\text{Region I } (\rho < a) : \lim_{b \rightarrow \infty} -E_0 \frac{4\chi b^2}{D_0} \rho \cos\theta = -\frac{4\chi}{(\chi+1)^2} E_0 \rho \cos\theta \rightarrow \text{uniform field different from external field}$$

$$\text{Region II } (\rho > a) : \lim_{b \rightarrow \infty} -E_0 \frac{2b^2}{D_0} \left[(\chi+1)\rho + (\chi-1)\frac{a^2}{\rho} \right] \cos\theta = \underbrace{-\frac{2}{\chi+1} E_0 \rho \cos\theta}_{\text{uniform field reduced by } \frac{2}{\chi+1} \text{ due to dielectric}} - \underbrace{\frac{2(\chi-1)}{(\chi+1)^2} \frac{a^2}{\rho} E_0 \cos\theta}_{\text{dipole term.}}$$

4.8 Two concentric conducting spheres of inner and outer radii a and b , respectively, carry charges $\pm Q$. The empty space between the spheres is half-filled by a hemispherical shell of dielectric (of dielectric constant ϵ/ϵ_0), as shown in the figure.



Problem 4.8

- (a) Find the electric field everywhere between the spheres.
- (b) Calculate the surface-charge distribution on the inner sphere.
- (c) Calculate the polarization-charge density induced on the surface of the dielectric at $r = a$.

We have 2 concentric conducting spheres, each must be an **equipotential surface**.

To satisfy this such that $E_t = 0$ everywhere, we must have $\vec{E}(r) = E(r) \hat{r}$ where $\vec{E}$ is normal to the surface.

a) Electric field between spheres.

Let $\vec{D} = \epsilon_0 \vec{E}$ (in vacuum) & $\vec{D}_R = \epsilon \vec{E}$ (in dielectric)

There is no free charge in region $a < r < b$. All free charge sits on conductor.

$$\therefore \nabla \cdot \vec{D} = 0 \Rightarrow \oint \vec{D} \cdot d\vec{a} = Q_{\text{free, enclosed}} \quad (\text{Gauss's Law})$$

$(a < r < b)$

Drawing a spherical Gaussian Surface just outside the inner sphere so that encloses $-Q$.

We have that :

$$\oint \vec{D} \cdot d\vec{a} = -Q$$

$$|\vec{D}_L| (2\pi r^2) + |\vec{D}_R| (2\pi r^2) = -Q \quad \vec{D} \text{ and } d\vec{a} \text{ are in the same direction.}$$

$$[\epsilon_0 E + \epsilon E] (2\pi r^2) = -Q$$

$$E(r) = -\frac{Q}{2\pi r^2 (\epsilon + \epsilon_0)} \quad \text{minus sign} \rightarrow \text{radially inward.}$$

Therefore, everywhere between the spheres $a < r < b$:

$$\vec{E}(r) = -\frac{Q}{2\pi r^2 (\epsilon + \epsilon_0)} \hat{r}$$

magnitude of E same between both material
magnitude of $\vec{D}$ differs between both material

b) We can relate the surface charge density σ_a of inner sphere by :

$$(\vec{D}_{out} - \vec{D}_{in}) \cdot \hat{n} = \sigma_a. \quad \vec{D}_{in} = 0 \text{ since } \vec{E} = 0 \text{ inside a conductor}$$

$$\left[\vec{D}_{out} \cdot \hat{r} = \sigma_a \right]_{on S}$$

We now find σ_a on the different hemispheres with different $\vec{D}$.

$$\sigma_a^{(dielectric)} = \vec{D}_R \cdot \hat{r} \Big|_{r=a} = -\epsilon E(a) = -\frac{\epsilon Q}{2\pi a^2 (\epsilon + \epsilon_0)}$$

$$\sigma_a^{(vacuum)} = \vec{D}_L \cdot \hat{r} \Big|_{r=a} = -\epsilon_0 E(a) = -\frac{\epsilon_0 Q}{2\pi a^2 (\epsilon + \epsilon_0)}$$

σ_a is uniform across hemispheres but different between them.

c) The polarization-charge density on surface of dielectric $r=a$ is given by :

$$\begin{aligned} \sigma_{pol} &= -[\vec{P}_{out} - \vec{P}_{in}] \cdot \hat{n}_{in \rightarrow out} \\ &= -\vec{P}_{out} \cdot \hat{r} \Big|_{r=a} \\ &= -P_{out}(r=a) \end{aligned}$$

$\vec{P}_{in} = 0$ since no fields in.

For linear materials, $\vec{P} = \epsilon_0 \chi_e \vec{E} = (\epsilon - \epsilon_0) \vec{E}$

$$\begin{aligned} \sigma_{pol} &= -P_{out}(r=a) = (\epsilon_0 - \epsilon) E_r(a) \\ &= \frac{(\epsilon - \epsilon_0) Q}{2\pi (\epsilon + \epsilon_0) a^2} \end{aligned}$$

→ uniform over dielectric hemisphere
0 over vacuum hemisphere.

* Interpretation : Negative charged inner conductor attracts positive charge onto dielectric surface.